Conceptual Design of Aircraft – propulsion

1. Propellers

2. Jet engines

1. Propellers – objectives:

By the end of the propellers session, the student will be able to:

1. Explain the principles of propellers operation, including effect of flight speed and ways to increase propeller thrust as blade twisting, blade sweeping-back and variable pitch.
2. Carry computations using the thin actuator disk model
3. Use propeller operational charts to calculate performance.

Introduction:

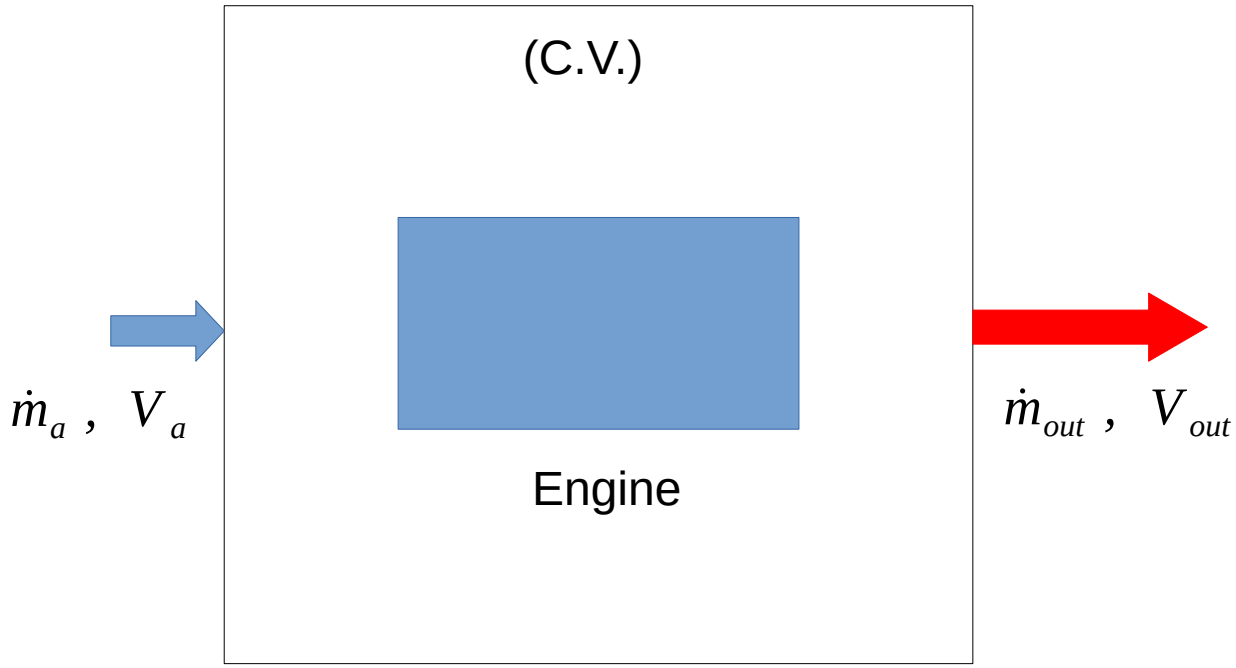
Both the propeller (using piston or gas turbine engine) and jet engines are air breathing engines (however exceptions as electric prolusion using propellers are emerging).

Air-breathing engine means that the oxygen needed to burn the fuel is taken from the atmosphere (unlike rockets that carry their own oxygen).

Simple analysis – propellers and jet engines

Whether the engine uses a propeller to act on the air or is a jet engine, both operate on the same fundamental principle originating from Newton's third law. Apply force on the air by accelerating it through the engine, and thus experience a force acting back from the air. This force is the thrust and this principle holds regardless how we get the power to the engine whether we burn fuel or get electrical power from a battery.

Simple thrust calculation (propeller and jet engine):
Treat the engine as a box, analyse from the point of view of a 'little person' standing on the engine and build a control volume (C.V.) around the engine.



$\dot{m}_a$ – air mass flow rate, V_a – flight velocity
$\dot{m}_{out}$ – gas mass flow rate leaving the engine
V_{out} – gas velocity (relative to the engine)

Continuity:
Propeller, we can take $\dot{m}_{out} = \dot{m}_a$

Jet engine, we can take $\dot{m}_{out} = \dot{m}_a + \dot{m}_f$
$\dot{m}_f -$ Fuel mass flow rate, fuel to air ratio $f = \dot{m}_f / \dot{m}_a$
However, $f \ll 1$ hence we can approxmiate even
for a jet engine $\dot{m}_{out} \simeq \dot{m}_a$

The static pressure of the incoming air is the ambient (atmospheric) pressure p_a. If the gas leaves the engine subsonically, i.e. $V_{out} < c_{out}$ (c is the local speed of sound), we can also approximate p_{out} as p_a (due to the theory of boundary layer). This surely happens for propellers where what leaves the control volume at V_{out} is the accelerated air. For jet engines, if the jet is supersonic then p_{out} may not be equal to p_a (in that case we will see a diamond structure inside the jet caused by a structure of shock waves and expansion fans), unless the jet is ideally expanded

In this module we will always assume $p_{out} = p_a$ in our calculations.



Example when
$p_{out} \neq p_a$

Therefore by doing a balance of force (Newton's second law) we get:

$$\dot{m}_a V_a + F = \dot{m}_a V_{out}$$

$$F = \dot{m}_a \left(V_{out} - V_a \right)$$

Where F is the uninstalled thrust (uninstalled means that we neglected the effect of the installation on the aircraft). Some books use the letter T for thrust.

Propulsive efficiency:

$$\eta_p = \frac{\text{Effective power}}{\text{Increase in flow power}}$$

$$\eta_p = \frac{F V_a}{\frac{\dot{m}_a V_{out}^2}{2} - \frac{\dot{m}_a V_a^2}{2}}$$

$$\eta_p = \frac{\dot{m}_a \left(V_{out} - V_a \right) V_a}{\frac{\dot{m}_a V_{out}^2}{2} - \frac{\dot{m}_a V_a^2}{2}}$$

$$\eta_p = \frac{2 V_a}{V_{out} + V_a}$$

Lets define $\Delta V = V_{out} - V_a$ hence

$$\eta_p = \frac{2V_a}{2V_a + \Delta V} = \frac{1}{1 + \frac{\Delta V}{2V_a}}$$

But

$$F = \dot{m}_a \left(V_{out} - V_a \right) = \dot{m}_a \Delta V$$

Hence

$$\eta_p = \frac{1}{1 + \frac{F}{2\dot{m}_a V_a}}$$

For the engine designer, the thrust F and aircraft velocity V_a are given by the aircraft manufacturer So the only variable that the engine designer can change in this simple analysis is the air mass flow rate dm_a/dt. The higher it is, the higher the propulsive efficiency η_p. Hence we see civil jet engines get bigger and bigger (it also reduces the jet noise). Of-course that cannot happen in supersonic flight, where a big engine will have a high wave drag penality.

The propeller engine has a very big air mass flow rate as compared to the jet engine, hence the attraction in using the propeller engine. It also naturally fits electric propulsion. However, the propeller considerably losses thrust at high speed flight and has been traditionally used for low speed flights as of cargo aircraft or short flights.

1. Propeller installation configurations

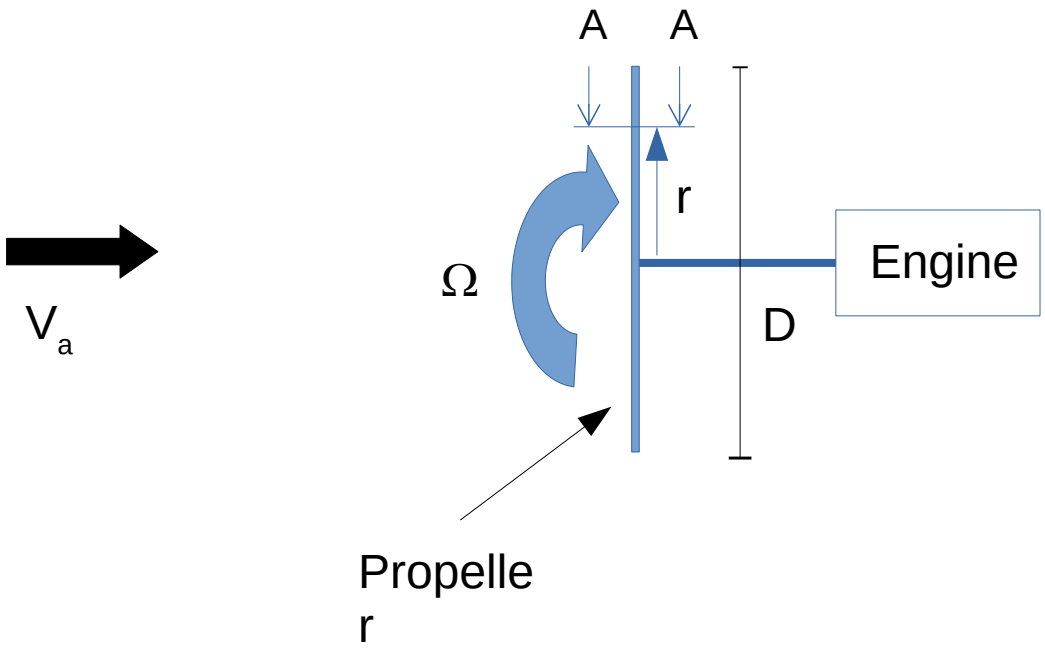


Puller configuration (C130 cargo aircraft)

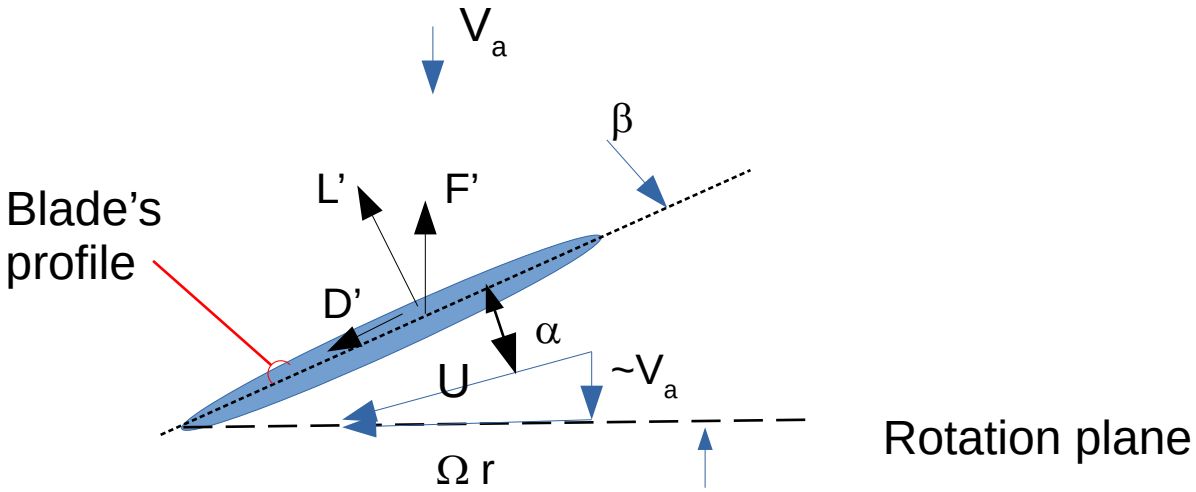


Pusher configuration (Piaggio P180)

Pitching the blade



Ω – propeller's angular speed (rad/s)
D – propeller's diameter
V_a – flight velocity



Section A-A

β(r) – pitch angle (rad), α – angle of attack (rad)
U – resultant velocity, $U^2 = (\Omega r)^2 + V_a^2$
L' – lift generated by profile (N/m)
D' - drag generated by profile (N/m)
F' - thrust generated by profile (N/m)

For high-performance propeller, the angle of attack α has to be high but less than the stall angle. Note that $\alpha < \beta$, where β is the pitch angle.

Near the hub (small r) U will be tilted more vertically since Ω r is small, so β has to be high in order to have a positive α. Near the blade's tip U will be shallower since Ω r will be large and thus β has to be small in order for the blade's profile not to stall (check it by sketching).

So the blade is twisted with a high pitch angle near the hub and a low pitch angle near the tip.

As the flight velocity (V_a) increases the angle of attack α will decrease if β and Ω are kept constant. Hence the thrust will decrease with V_a.

The decrease of the thrust with the increase of V_a can be partly mitigated by:
1. Have a variable pitch propeller, hence increase β as V_a increases. This is a partial remedy since it will still have L' tilting away from the (opposite) direction of V_a and hence F' will still decrease (check it by sketching).
2. Increase Ω, but this can cause the tip to go supersonic and thus introduces a high wave drag due to shock waves. The tip's wave drag can be partly mitigated by sweeping back the blade's tip.

Airbus A400M, note the swept back blades



Blackhawk UH-60, note the sweptback tip of the blade, to avoid normal shock-waves due to the supersonic speed of the blade's tip.



To increase propeller's efficiency, we may like to increase its diameter D, so more air can pass through the propeller and thus the propulsive efficiency η_p is increased for the same thrust.

However, there are practical limitations on D:
1. Ground clearance.
2. Structural stress levels.
3. Compressibility effects (wave drag at the tip).
4. Noise.

For straight blades, we want to keep the blade's tip (*r=R=D/2*) subsonic.

$$U_{tip}^2 = \Omega^2 R^2 + V_a^2$$

$$\Omega = 2\pi n, \text{ n is frequency (Hz)}$$

$$U_{tip}^2 = \pi^2 n^2 D^2 + V_a^2$$

$$M_{tip}^2 = \frac{\pi^2 n^2 D^2}{c^2} + M_a^2, \text{ c is the speed of sound (m/s)}$$

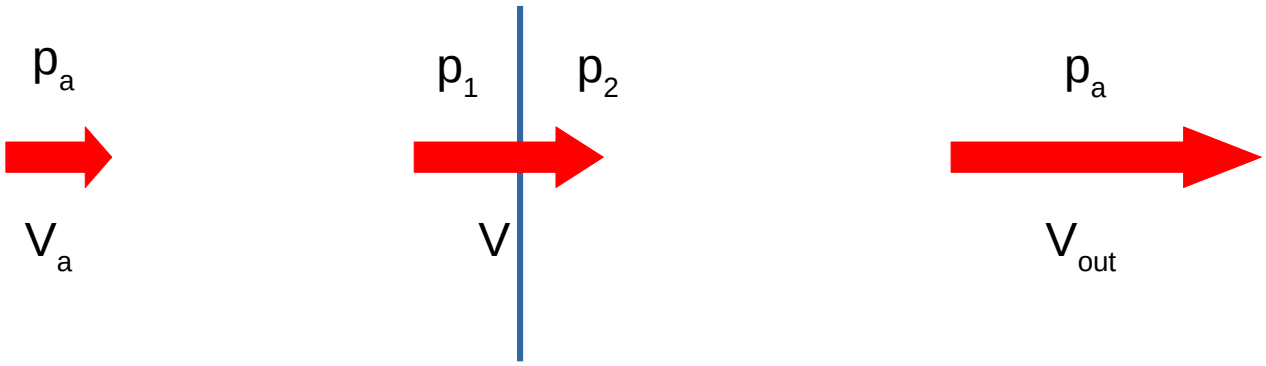
$$D = \frac{c}{\pi n}\sqrt{M_{tip}^2 - M_a^2}$$

$$M_{tip} \leqslant 0.72, \text{ for regular propellers}$$

Simple momentum theory for propellers
Thin actuator disc model

The propeller is modelled as a thin porous disc.

Assumptions:
1. Incompressible flow.
2. Potential flow in front and after the disc (but not inside the disc).
3. There is a pressure jump over the disc but the velocity through the disc is continuous.
4. No swirl in the wake.



$$F = (p_2 - p_1) A \ , \quad A = \frac{\pi D^2}{4} \ , \quad \text{D is the diameter of the disc}$$

$$F = \dot{m}(V_{out} - V_a) = \rho V A (V_{out} - V_a)$$

Bernoulli equation can be used in front of the disc and after the disc, but not through the disc.

$$p_a + \frac{\rho V_a^2}{2} = p_1 + \frac{\rho V^2}{2}$$

$$p_a + \frac{\rho V_{out}^2}{2} = p_2 + \frac{\rho V^2}{2}$$

$$p_2 - p_1 = \frac{\rho\left(V_{out}^2 - V_a^2\right)}{2}$$

So

$$F = \left(p_2 - p_1\right) A = \frac{\rho\left(V_{out}^2 - V_a^2\right) A}{2}$$

On the other hand from momentum balance, the thrust F is also

$$F = \rho V A\left(V_{out} - V_a\right)$$

Both expressions for the thrust F must be the same leading to:

$$V = \frac{V_{out} + V_a}{2}, \quad F = \frac{\rho A\left(V_{out}^2 - V_a^2\right)}{2}$$

This result is not complete, because we still do not know what is inside the disc, i.e. the blades' aerodynamic performance. This is accounted in the blade element momentum (BEM).

Example:

A propeller is tested at sea-level condition of $\rho = 1.225\ kg/m^3$ on a stationary rig

Data:

$D = 1\ m$, $V_{out} = 20\ m/s$

$c = 340\ m/s$

Question:

$F = ?$ $\quad p_2 - p_1 = ?$

$\underline{P} = ?$ $\quad \underline{P}$ is power

$\eta_p = ?$

$M_{tip} = ?$ $\quad$ If $n = \frac{\Omega}{2\pi} = 40\ Hz$

Solution:

$$F = \dot{m}(V_{out} - V_a) = \rho V A (V_{out} - V_a)$$

$$V = \frac{V_{out} + V_a}{2} = \frac{20+0}{2} = 10\ m/s$$

$$F = 1.225 \cdot 10 \cdot \frac{\pi \cdot 1}{4} (20 - 0) = 192.42\ N$$

$$F = (p_2 - p_1) A$$

$$192.42 = (p_2 - p_1) \frac{\pi \cdot 1}{4}$$

$\hookrightarrow$

$$p_2 - p_1 = 245\ Pa$$

$$\underline{P} \cong F \cdot V = 192.42 \cdot 10 = 1924.2\ W$$

$\eta_p = 0$ because $V_a = 0$.

$$V_{tip}^2 = \pi^2 n^2 b^2 + V^2$$

$$= (\pi \cdot 40 \cdot 0.1)^2 + 10^2$$

$V_{tip} = 126$ m/s

$M_{tip} = \frac{V_{tip}}{c} = \frac{126}{340} = 0,371$

Propeller activity factor - AF

The power P is related to the propeller's torque:

$$P = \Omega Q$$

where the frequency n is $n = \frac{\Omega}{2\pi}$

The torque Q is related to the drag acting on the blades. Hence the propeller activity factor can be defined as:

$$P = 4 B \rho \pi^3 n^3 \bar{c}_d \left(\frac{D}{10} \right)^5 AF$$

Where

B - number of blades

$\bar{c}_d$ - blade average drag coefficient

Typical values for AF: 70<AF<200
AF=162 for the C130 cargo aircraft

Define the following coefficients:

$$C_F = \frac{F}{\rho n^2 D^4}, \text{ propeller thrust coefficent}$$

$$C_Q = \frac{Q}{\rho n^2 D^5}, \text{ propeller torque coefficent}$$

$$C_P = \frac{P}{\rho n^3 D^5}, \text{ propeller power coefficent}$$

Once the activity factor AF is given, C_F or C_P for static condition (i.e. V_a=0), can be found using the propeller performance chart as illustrated in the next slide.

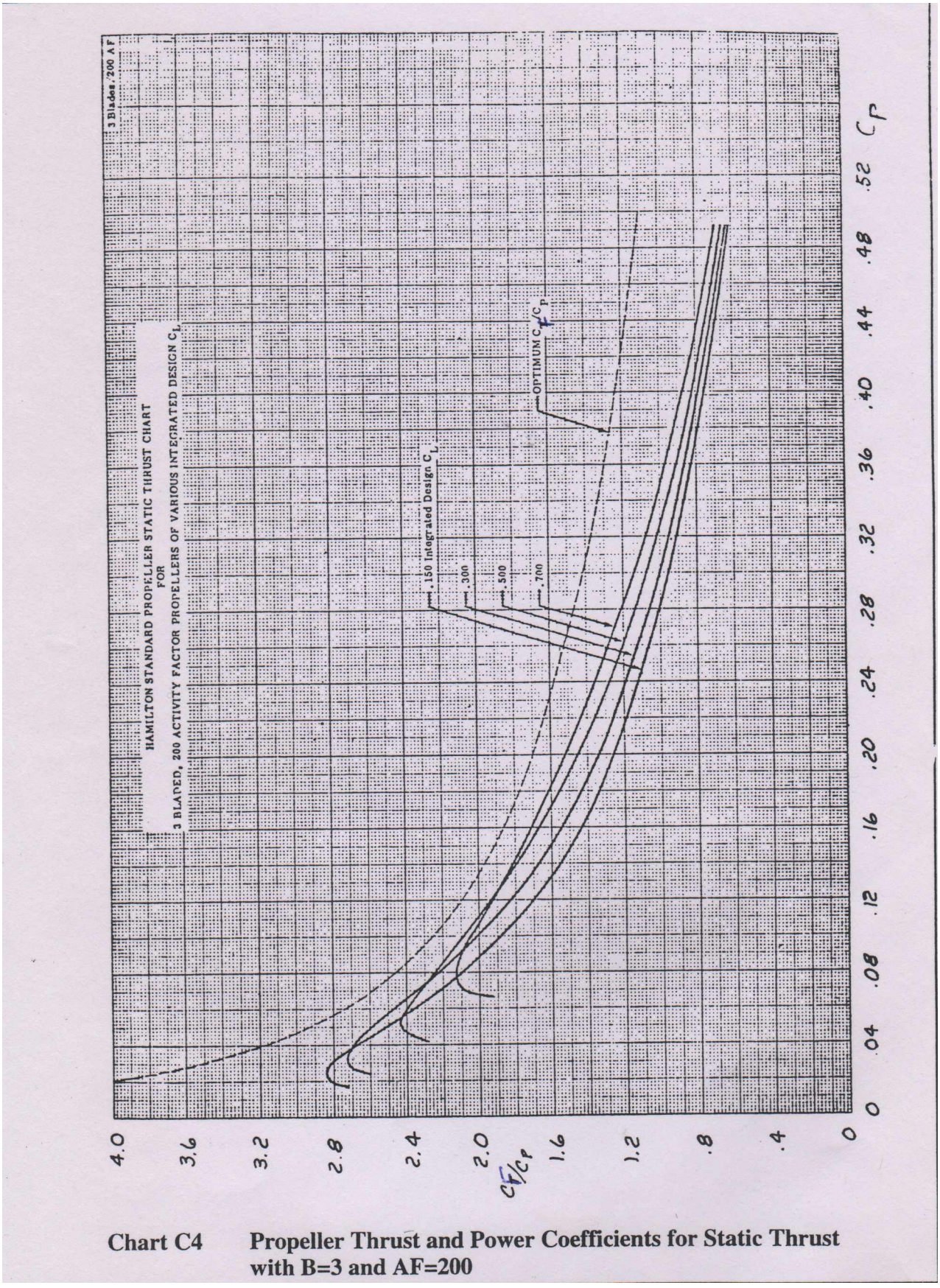


Chart C4 **Propeller Thrust and Power Coefficients for Static Thrust with B=3 and AF=200**

From the module textbook “Airplane aerodynamics and performance” by Lan & Roskam

Approximate method to calculate thrust at flight

Define propeller advance ratio:

$$J = \frac{V_a}{n D}$$

Hence

$$\eta_p = \frac{F V_a}{P} = J \frac{C_F}{C_P}$$

For an ideal propeller we can assume that P and C_P do not depend on V_a. Hence, once we found P or C_P for the static condition we can use those values for the flight condition. The thrust or its coefficient C_F can be calculated if the propulsive efficiency η_p is known. η_p can be found from the following performance chart (the blade's pitch has to be given) or is given.

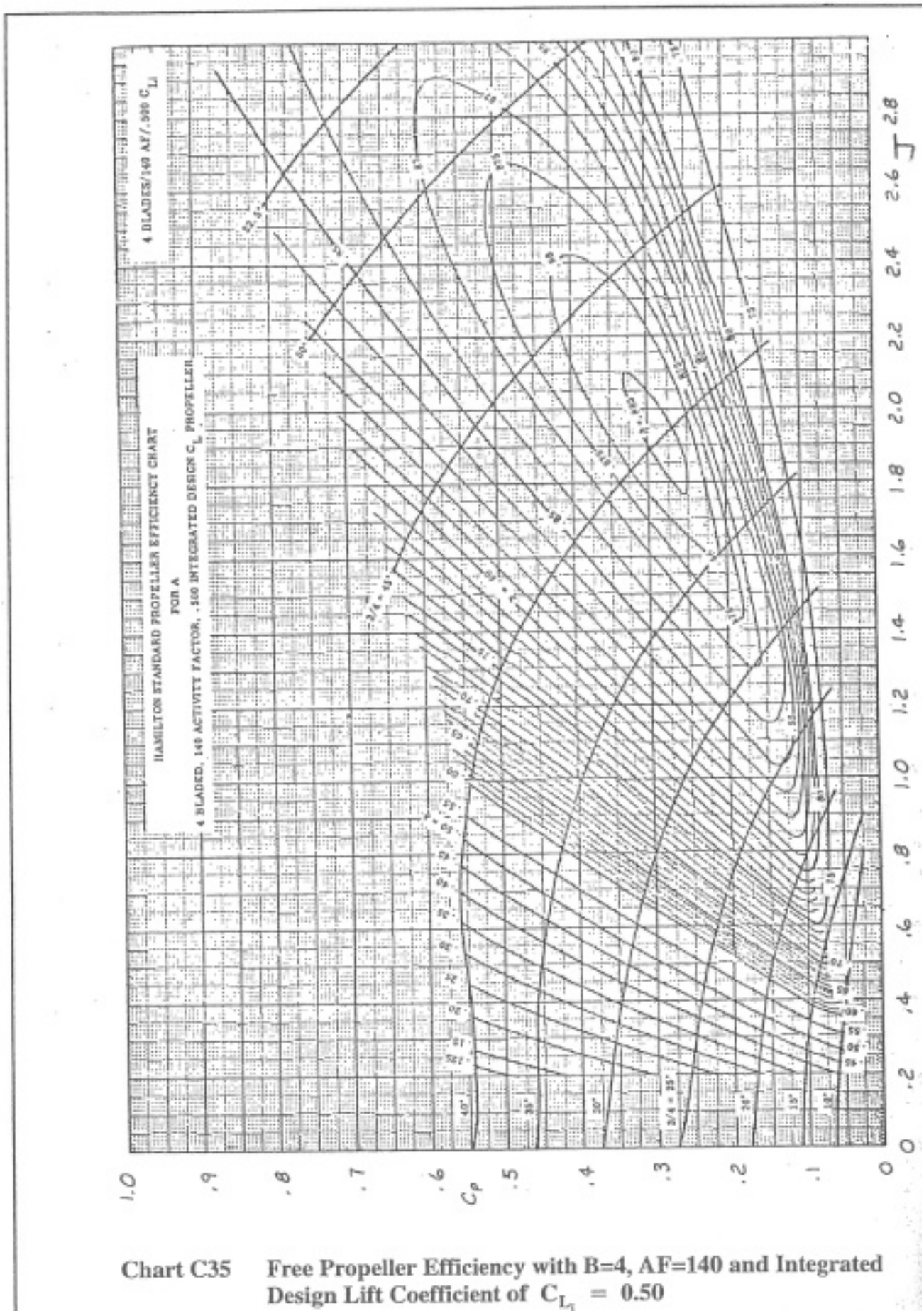


Chart C35 Free Propeller Efficiency with B=4, AF=140 and Integrated Design Lift Coefficient of C_{L_i} = 0.50

From the module textbook “Airplane aerodynamics and performance” by Lan & Roskam

Recent propeller and rotor developments

Future aircraft, Airbus bird of prey concept



Complex aerodynamic design

Airbus city taxi concept



Coaxial rotors, new design, well fitted for electric propulsion

Further information on coaxial rotors can be found in Avital, Korakianitis and Motallebi (2019) Low Reynolds number Proprotor aerodynamic performance improvement using the continuous surface curvature design approach, *Aero J* **123** (1259), 20-38

Effect of propeller installation

What we calculated until now was $\eta_{p,free}$, meaning we neglected the effect of the installation of the engine

$\eta_{p,installed} = F_{blockage} * F_{scrubbing} * F_{compressibility} * \eta_{p,free}$

$F_{blockage}$ depends on the inlet installation (see next slide for the configuration) and whether the propeller is a puller or a pusher.

$F_{scrubbing} = 1 - 1.558 * \sigma * f_{slip}/D^2$

$\sigma = \rho / \rho_0$

$F_{slip} = 0.004 * S_{wet,\ slipstream}$

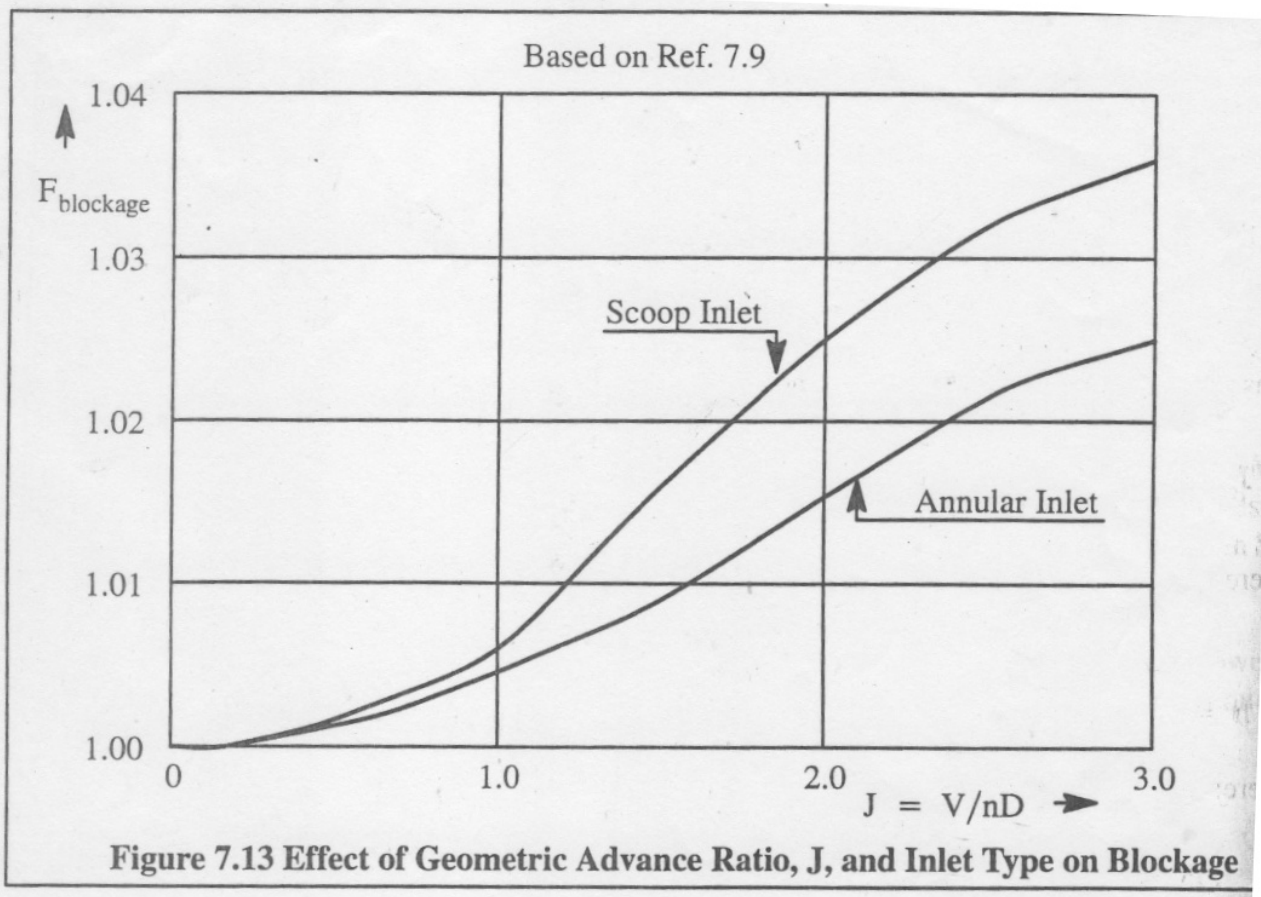


Figure 7.13 Effect of Geometric Advance Ratio, J, and Inlet Type on Blockage

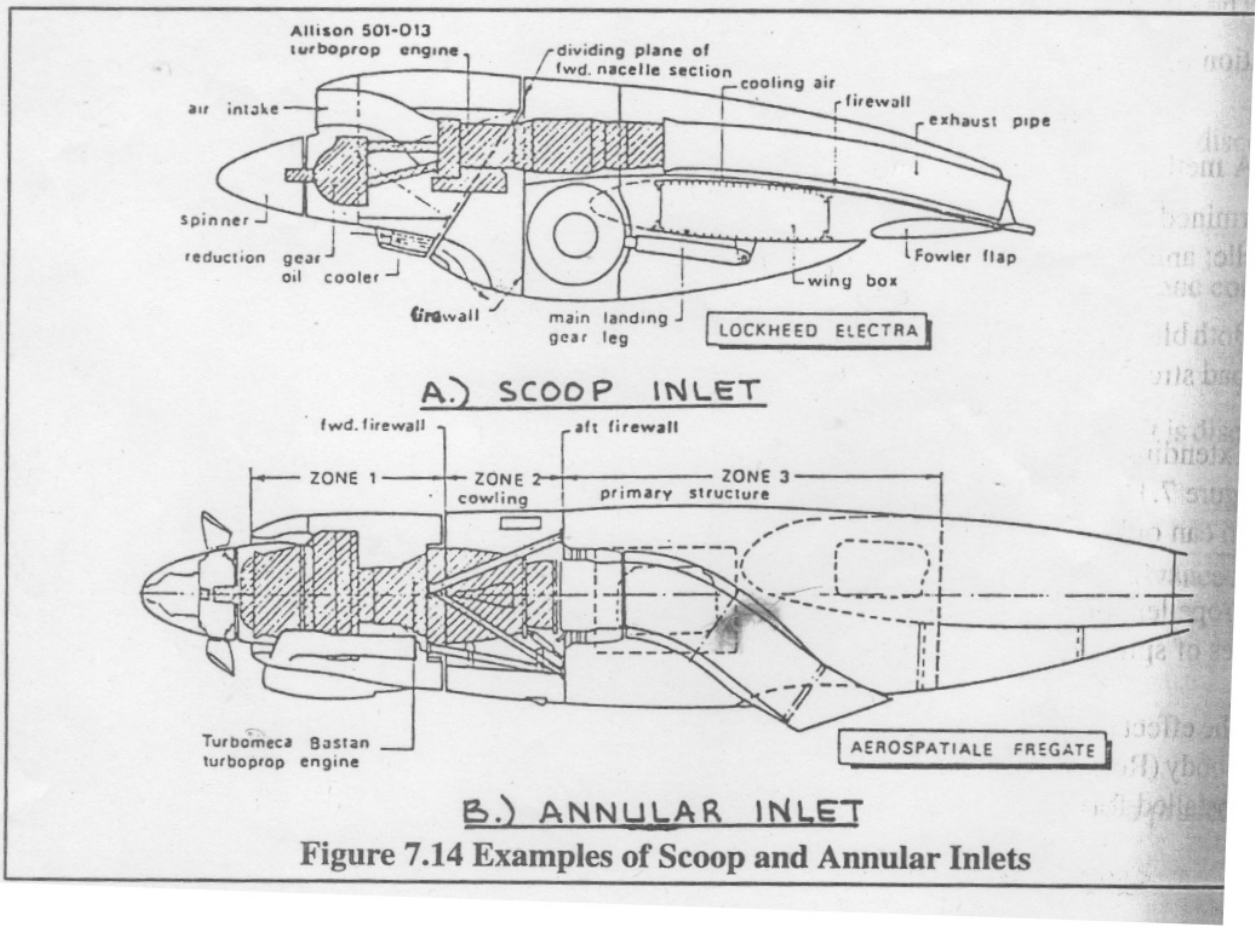


Figure 7.14 Examples of Scoop and Annular Inlets

From the module textbook “Airplane aerodynamics and performance” by Lan & Roskam

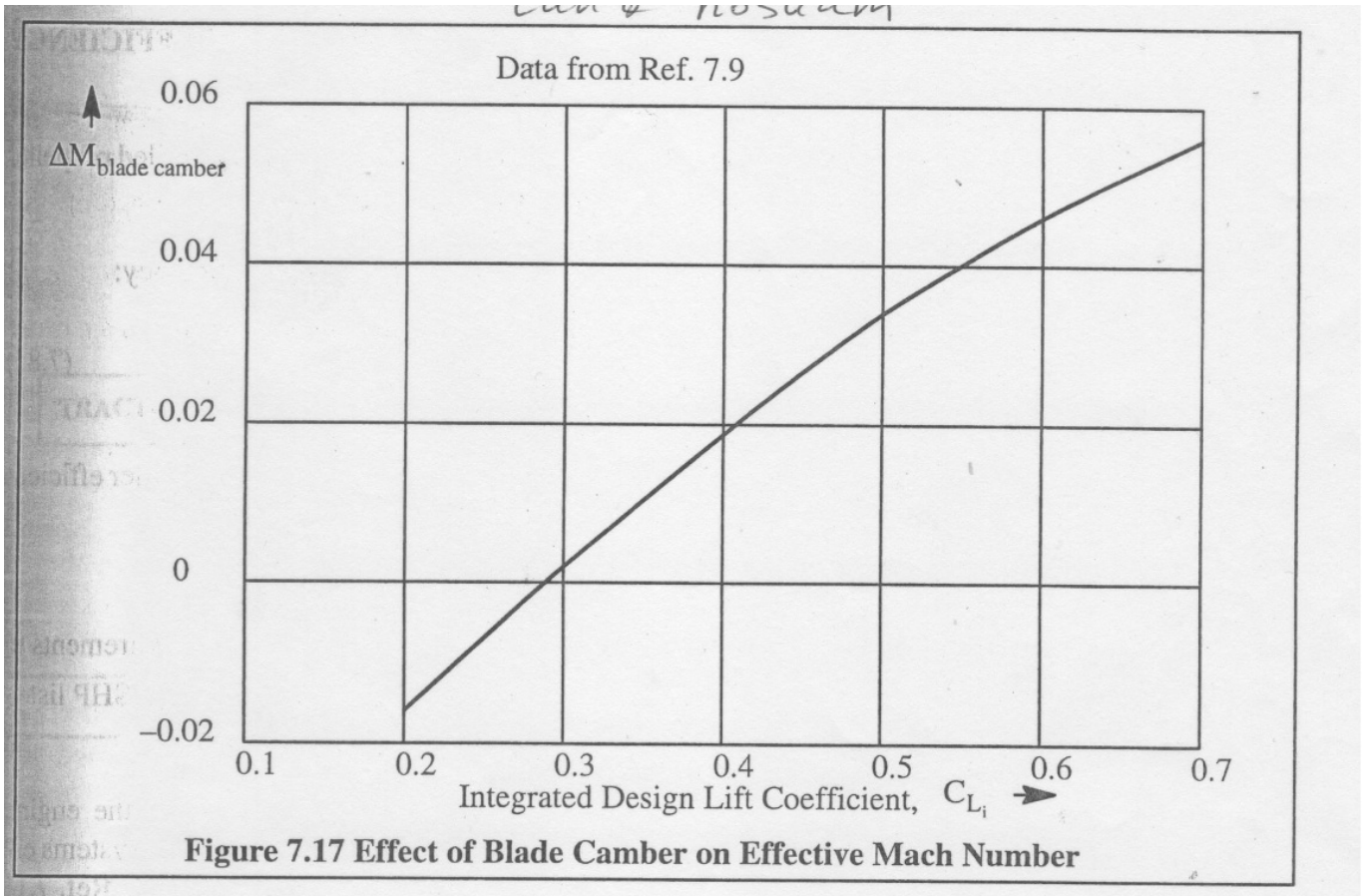


Figure 7.17 Effect of Blade Camber on Effective Mach Number

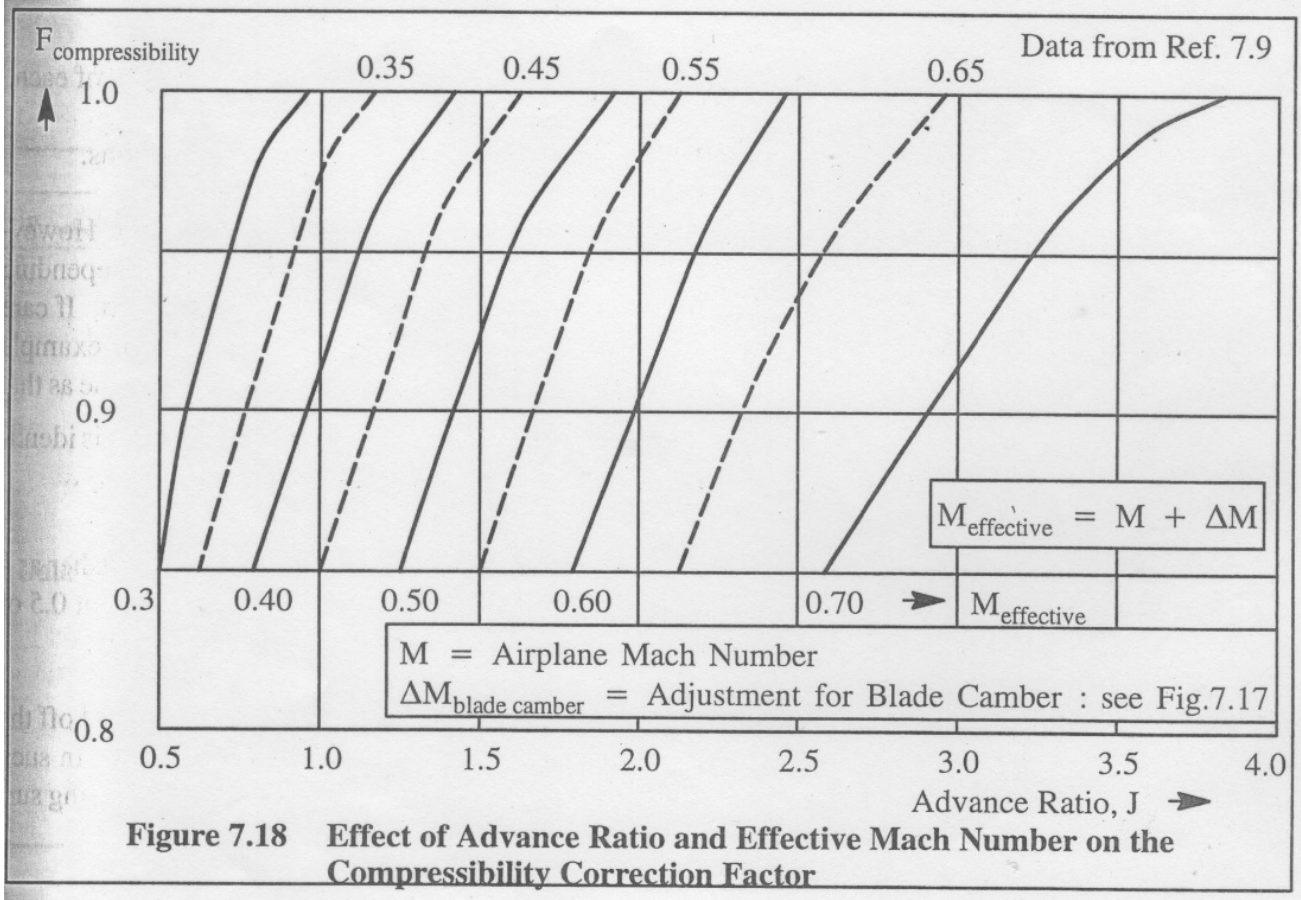


Figure 7.18 Effect of Advance Ratio and Effective Mach Number on the Compressibility Correction Factor

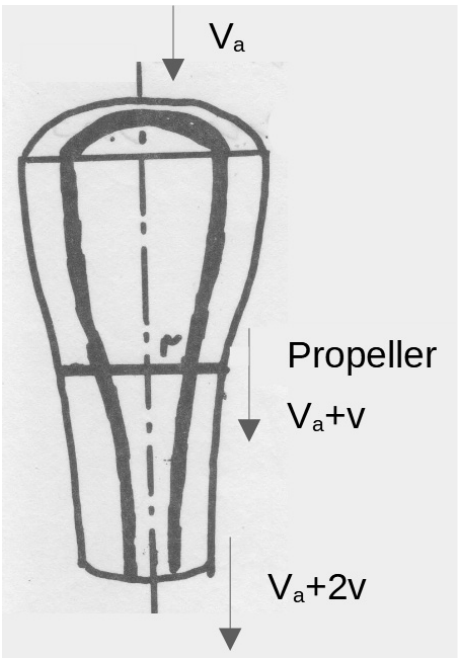
From the module textbook “Airplane aerodynamics and performance” by Lan & Roskam

Appendix:
Simplified blade element momentum (BEM) method

Define the velocity going the disk as $V(r) = V_a + v(r)$, where V_a is the flight speed. Then by control volume arguments but for a ring control volume one can get that the thrust dF(r) per a ring of the propeller's disk is:

$$dF = 4\rho\pi r\left[V_a + v(r)\right]v(r)\,dr$$

$$F = 4\rho\pi\int_0^R r\left[V_a + v(r)\right]v(r)\,dr$$



This can be symbolically rewritten as:

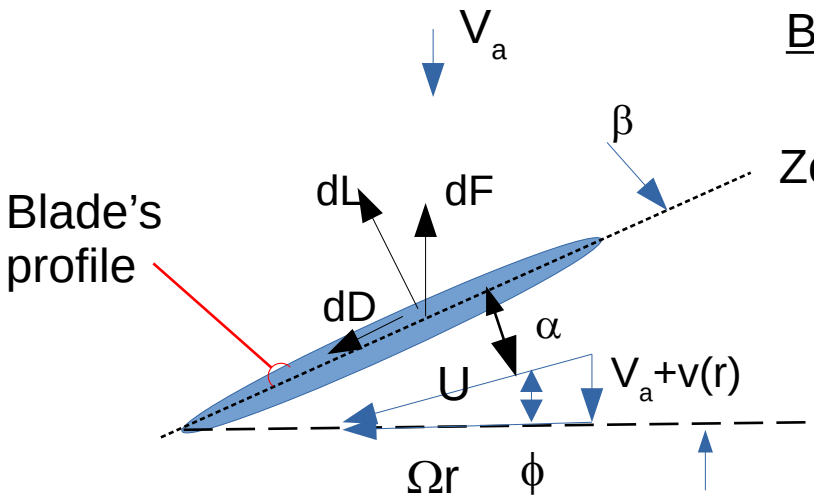
$$F = 2\rho A\left(V_a + k_1 v_{0.75}\right)k_2 v_{0.75}$$

Where $A = \pi R^2$ is the propeller disk area
and F is the thrust produced by the propeller

$V_{0.75}$ = v(r=0.75R)

If we assume a uniform distribution of v(r), then $k_1 = k_2 = 1$

If we assume a linear distribution of v(r), then $k_1 = 1$ and $k_2 = 0.888$



By blade element method

$$\tan\phi = \frac{V_a + v(r)}{\Omega r}$$

We assume $\phi \ll 1$ rad, hence

$$\phi \simeq \frac{V_a + v(r)}{\Omega r},\ U \simeq \Omega r$$

We also assume $dD/dL \ll 1$ hence $dF \simeq B * dL$

$$C_L = a(r)\alpha = a(r)\left[\beta(r) - \phi(r)\right]$$

$$dF = B * dL = \frac{B}{2}\rho U^2 c(r) C_L dr = \frac{B}{2}\rho(\Omega r)^2 c(r) a(r)\left[\beta(r) - \frac{V_a + v(r)}{\Omega r}\right]dr$$

$$F = \int_0^R dF = \frac{B}{2}\rho\Omega^2 \int_0^R a(r)c(r)r^2\left[\beta(r) - \frac{V_a + v(r)}{\Omega r}\right]dr$$

Where B is the number of blades. If two calculations of the thrust F by the momentum theory and the blade element method are correct, both expressions for dF (and F) must be the same!

This will give an equation for v(r) that has to be solved for various points of r along the blade.

If we assume a uniform a(r), c(r), β(r) and v(r), we get:

$$F = \frac{B}{2}\rho\,\Omega\,R^2\,a\,c\left(\frac{\Omega R\beta}{3} - \frac{V_a + v_{0.75}}{2}\right)$$

Requiring this expression of F to be equal to the expression of F by the momentum theory while assuming uniform v yields an algebraic equation for $v_{0.75}$ that can be easily solved.

In this simplified BEM method we neglected the effect of induced tangential velocity at the propeller's disk that will affectively change $\Omega * r$ as seen by the blade.

In order to also account for the induced tangential velocity which can become important in a heavy loaded disk, the full BEM has to be solved, where expressions for the torque Q (or power P) of the propeller are also derived using the momentum theory and the blade element method. Then as for the thrust F, the expressions for the torque must also be made equal. This will give two equations, one for the induced axial velocity v(r) and another for the induced tangential velocity.

For further details, see "Theoretical Aerodynamics" by Milne-Thomson, "Rotary-wing aerodynamics" by Stepniewski and Keys.

Current freeware to design and analyse propellers (google them): Javaprop, prop_design, Xrotor

Another method to calculate thrust and power is to use the propeller's performance charts.